

Project Name: Automating CI/CD Pipeline for Spring Boot Application Deployment on AWS

Objective: To implement a CI/CD pipeline using AWS services for automating the deployment of a Spring Boot application on Amazon ECS with Docker, integrating CodePipeline, CodeBuild and ECR for seamless updates

Problem Statement: Imagine a retail company that has a Spring Boot-based inventory management system. The system is used by several departments and is hosted in containers to ensure scalability. The company wants to reduce the manual intervention required to deploy updates by automating the build and deployment process. By setting up a CI/CD pipeline, developers can push updates to a GitHub repository, automatically triggering a sequence of actions that build, containerize, and deploy the application without downtime. This ensures faster and more reliable system updates, minimizing operational delays.

Steps:

1. will create a cicd pipeline via code pipeline where we will deploy a springboot application on ECS docker containers
2. Process:
 - a. Our code would be picked up
 - b. it will go to code build where the project will be converted to the war file (single executable file)
 - c. we execute the dockerfile to create custom image which would be pushed to ecr
 - d. from there it would be picked by ecs task to deploy the project

AWS Springboot Application Repo: <https://github.com/pradeepviswa/aws-devops-springbootapp>

 **aws-devops-springbootapp** Public

forked from [akshu20791/aws-devops-springbootapp](#) Pin Watch 0

master 1 Branch 0 Tags Add file Code

This branch is up to date with [akshu20791/aws-devops-springbootapp:master](#). Contribute Sync fork

 **akshu20791** Update buildspec.yml

8cc1d7b · 2 weeks ago 12 Commits

 .mvn/wrapper	my first commit	2 years ago
 src	my first commit	2 years ago
 .gitignore	my first commit	2 years ago
 Dockerfile	my first commit	2 years ago
 README.md	Update README.md	2 years ago
 buildspec.yml	Update buildspec.yml	2 weeks ago
 mvnw	my first commit	2 years ago
 mvnw.cmd	my first commit	2 years ago
 pom.xml	my first commit	2 years ago

1. Create ECR

Amazon ECR > Private registry > Repositories > Create private repository

Create private repository

General settings

Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.

612990353408.dkr.ecr.us-east-1.amazonaws.com/

9 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters ...

Image tag settings [Info](#)

Image tag mutability
Choose the tag mutability setting.

☒ **Mutable**
Image tags can be overwritten.

☐ **Immutable**
Image tags can't be overwritten.

- ## 2. View push commands and copy command under “Retrieve an authentication token and authenticate your Docker client to your registry. Use the AWS CLI:”

Private repositories (1)

Search by repository substring

Repository name	URI	Created at	Tag immutability	Encryption type
awsdevops	612990353408.dkr.ecr.us-east-1.amazonaws.com/awsdevops	May 04, 2026, 15:39:34 (UTC+05.5)	Mutable	AES-256

Command: `aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 612990353408.dkr.ecr.us-east-1.amazonaws.com`

Push commands for awsdevops

macOS / LinuxWindows

Make sure that you have the latest version of the AWS CLI and Docker installed. For more information, see [Getting Started with Amazon ECR](#).

Use the following steps to authenticate and push an image to your repository. For additional registry authentication methods, including the Amazon ECR credential helper, see [Registry Authentication](#).

- Retrieve an authentication token and authenticate your Docker client to your registry. Use the AWS CLI:

```
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 612990353408.dkr.ecr.us-east-1.amazonaws.com
```

Note: If you receive an error using the AWS CLI, make sure that you have the latest version of the AWS CLI and Docker installed.
- Build your Docker image using the following command. For information on building a Docker file from scratch see the instructions [here](#). You can skip this step if your image is already built:

```
docker build -t awsdevops .
```
- After the build completes, tag your image so you can push the image to this repository:

```
docker tag awsdevops:latest 612990353408.dkr.ecr.us-east-1.amazonaws.com/awsdevops:latest
```
- Run the following command to push this image to your newly created AWS repository:

```
docker push 612990353408.dkr.ecr.us-east-1.amazonaws.com/awsdevops:latest
```

Close

3. Go to github repo: <https://github.com/pradeepviswa/aws-devops-springbootapp.git>
Click on buildspec.yml -> edit the file (by clicking in pencil icon)
Update line no 10 with the code copied above

aws-devops-springbootapp / buildspec.yml in master

Edit Preview

```
1 version: 0.2
2 phases:
3   pre_build:
4     commands:
5       - echo Logging in to Amazon ECR...
6       #checking version
7       - aws --version
8       #- $(aws ecr get-login --region ap-south-1 --no-include-email)
9       # update below
10      - aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 612990353408.dkr.ecr.us-east-1.amazonaws.com
11      - # Replace with this to your repository URI
```

4. Go back to ECR and copy ECR Image URI
URI: `612990353408.dkr.ecr.us-east-1.amazonaws.com/awsdevops`

Private repositories (1)

Search by repository substring

Repository name	Repository URI copied	Created at
awsdevops	612990353408.dkr.ecr.us-east-1.amazonaws.com/awsdevops	May 04, 2026, 15:39:34 (UTC+05.5)

5. Paste in buildspec.yml file in github account
On line 12 of buildspec.yml file.

aws-devops-springbootapp / buildspec.yml in master

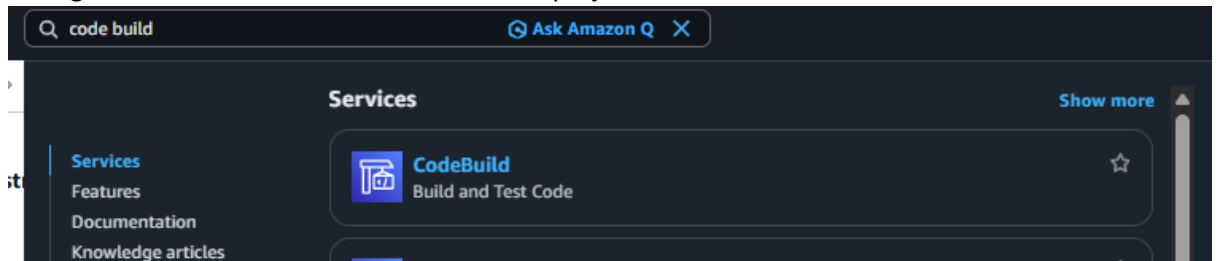
Edit Preview

```
1 version: 0.2
2 phases:
3   pre_build:
4     commands:
5       - echo Logging in to Amazon ECR...
6       #checking version
7       - aws --version
8       #- $(aws ecr get-login --region ap-south-1 --no-include-email)
9       # update below
10      - aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 612990353408.dkr.ecr.us-east-1.amazonaws.com
11      - # Replace with this to your repository URI
12      - REPOSITORY_URI=612990353408.dkr.ecr.us-east-1.amazonaws.com/awsdevops
13      - IMAGE_TAG=build-${echo $CODEBUILD_BUILD_ID | awk -F":" '{print $2}'}
```

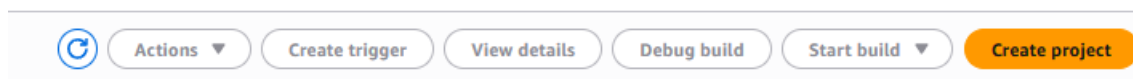
6. Build Commands in buildspec.yml file

```
14 build:
15   commands:
16     - echo Build started on `date`
17     - echo building the Jar file
18     - mvn clean install
19     - echo Building the Docker image...
20     - docker build -t $REPOSITORY_URI:latest .
21     - docker tag $REPOSITORY_URI:latest $REPOSITORY_URI:$IMAGE_TAG
```

7. Now go to code build and create a code build project



8. Create project



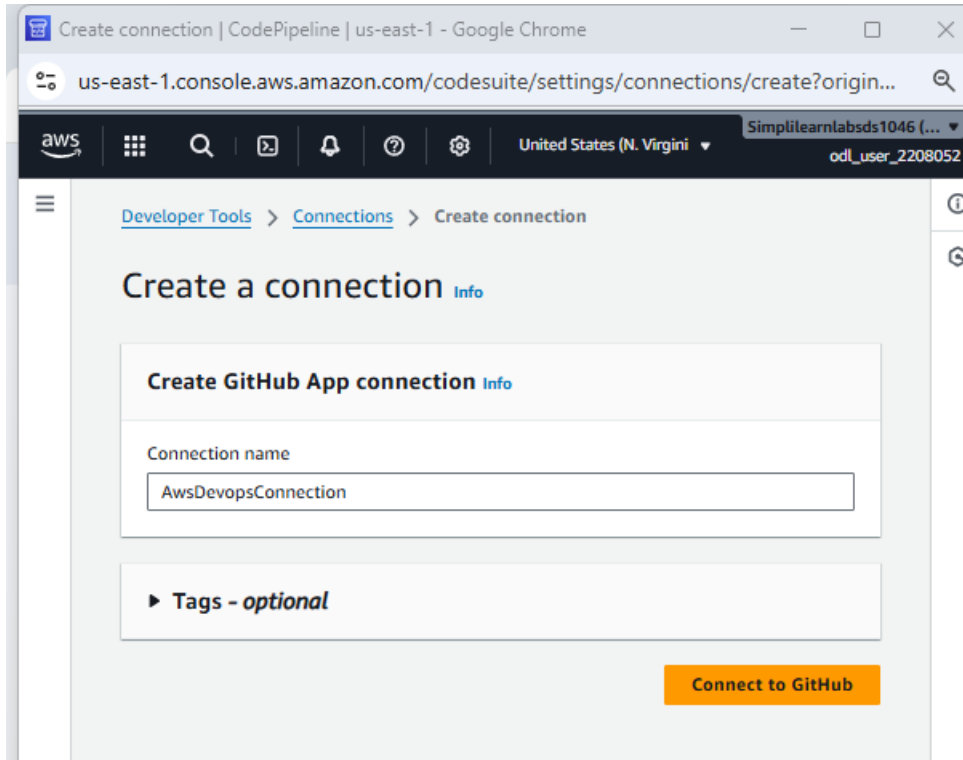
Name: AwsDevopsProject

A screenshot of the 'Create build project' form in the AWS CodeBuild console. The form has a title 'Create build project' and a section 'Project configuration' with a 'Project name' field containing 'AwsDevopsProject'. Below this is a 'Source' section with 'Source 1 - Primary' and 'Source provider' set to 'GitHub'. A 'Credential' section shows a red error message: 'You have not connected to GitHub. Manage account credentials.'.

9. Click on manage default source credentials

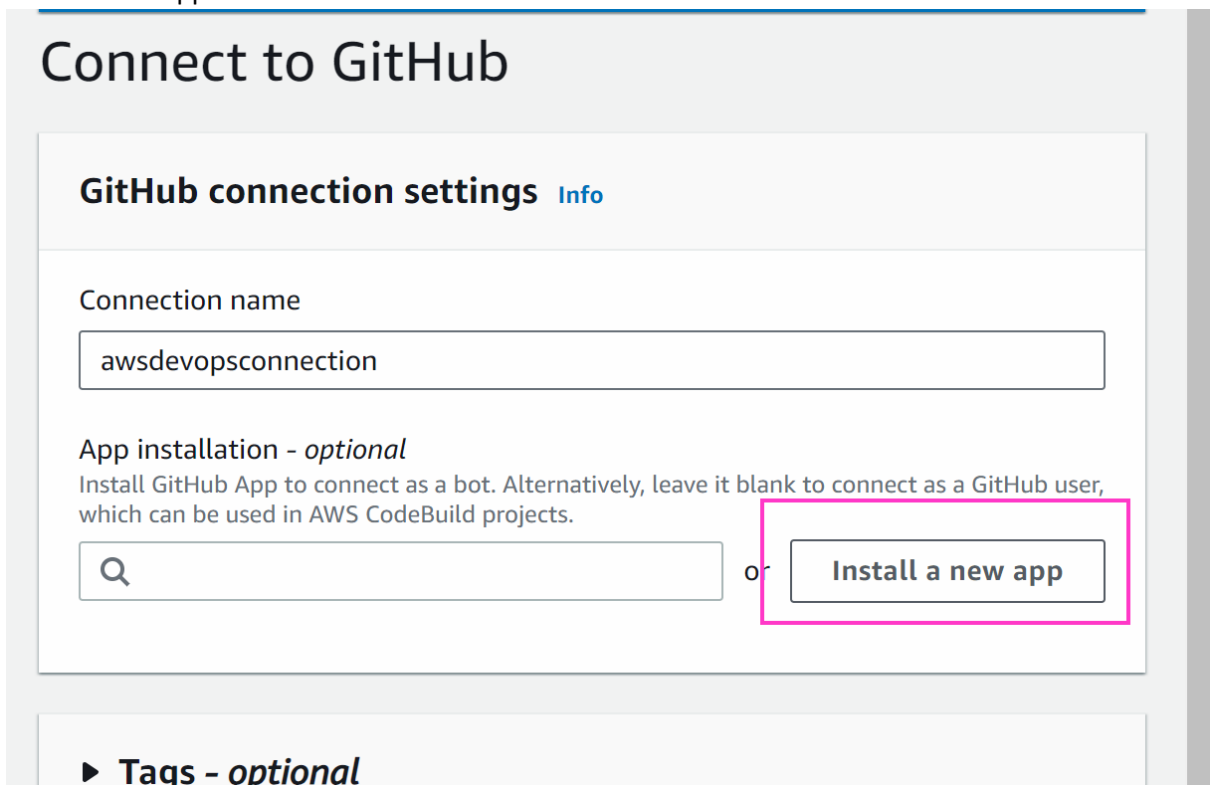
A screenshot of the 'Manage default source credential' form in the AWS CodeBuild console. The form has a title 'Manage default source credential' and a 'Source Provider' dropdown set to 'GitHub'. Below this is a 'Credential type' section with three options: 'GitHub App' (selected), 'Personal access token', and 'OAuth app'. The 'GitHub App' option is highlighted with a pink box. Below the 'Credential type' section is a 'Connection' section with a text input field and a 'Save' button. The text input field contains the text 'You can create a new GitHub connection by using an AWS managed GitHub App'.

10. A pop up will open



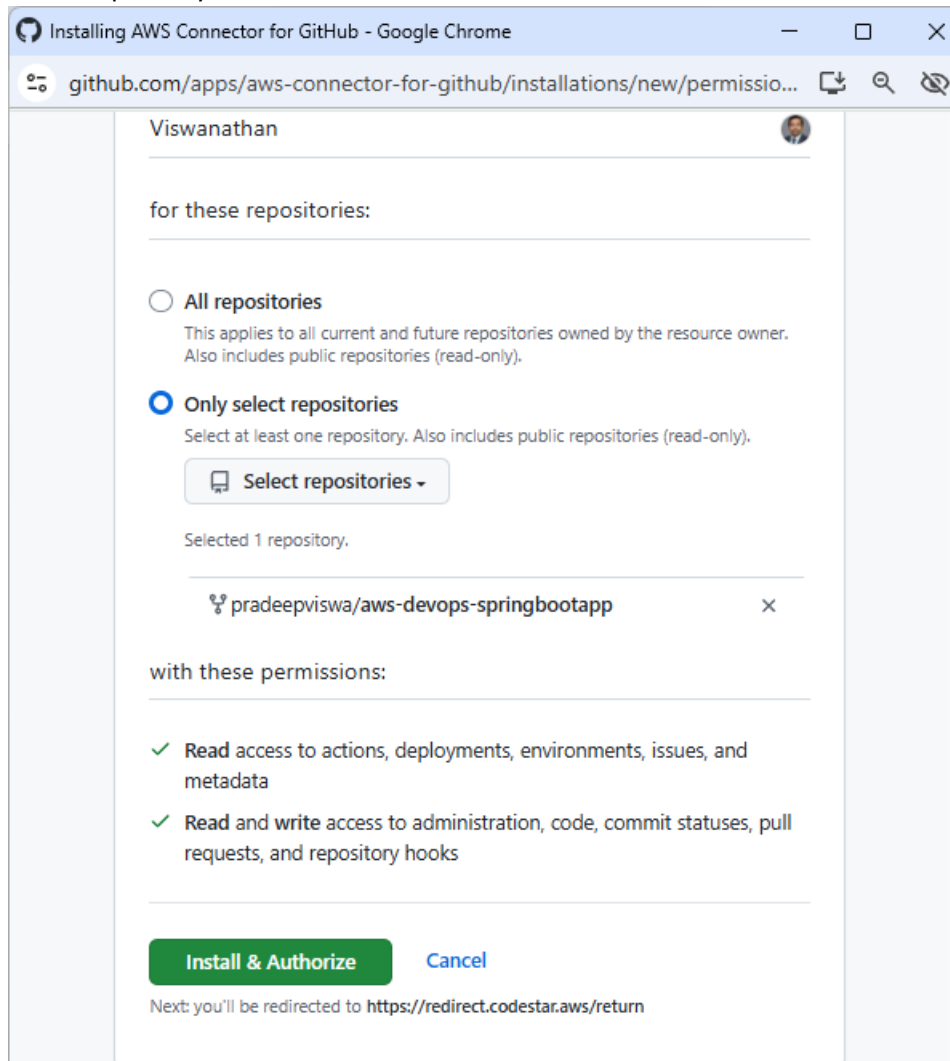
The screenshot shows the AWS CodePipeline console in a Google Chrome browser. The address bar shows the URL: `us-east-1.console.aws.amazon.com/codesuite/settings/connections/create?origin...`. The page title is "Create connection | CodePipeline | us-east-1 - Google Chrome". The breadcrumb navigation is "Developer Tools > Connections > Create connection". The main heading is "Create a connection [Info](#)". Below this is a section titled "Create GitHub App connection [Info](#)". It contains a "Connection name" input field with the value "AwsDevopsConnection". There is a "Tags - optional" section with a right-pointing triangle icon. At the bottom right is an orange button labeled "Connect to GitHub".

11. Install a new app



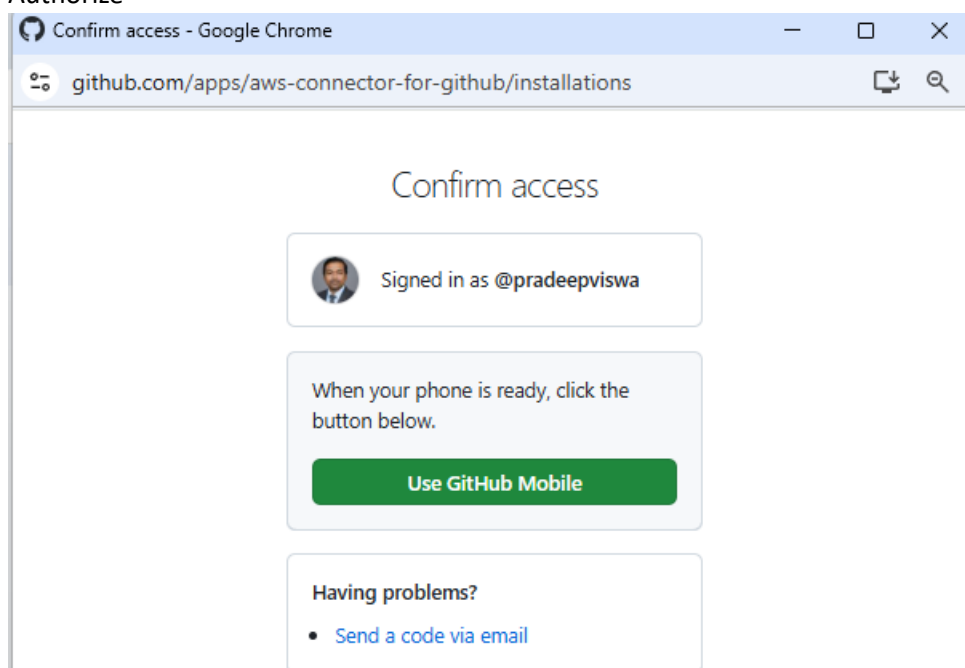
The screenshot shows a "Connect to GitHub" dialog box. The main heading is "Connect to GitHub". Below it is a section titled "GitHub connection settings [Info](#)". It contains a "Connection name" input field with the value "awsdevopsconnection". Below this is a section titled "App installation - optional" with the text: "Install GitHub App to connect as a bot. Alternatively, leave it blank to connect as a GitHub user, which can be used in AWS CodeBuild projects." There is a search input field with a magnifying glass icon. To the right of the search field is the word "or" and a button labeled "Install a new app" which is highlighted with a pink border. At the bottom is a section titled "Tags - optional" with a right-pointing triangle icon.

12. Select repository then click on **Install & Authorize** button



The screenshot shows a web browser window titled "Installing AWS Connector for GitHub - Google Chrome". The address bar shows the URL "github.com/apps/aws-connector-for-github/installations/new/permisso...". The page content is for user "Viswanathan". It asks for permissions "for these repositories:". There are two radio button options: "All repositories" (unselected) and "Only select repositories" (selected). Below the selected option, it says "Select at least one repository. Also includes public repositories (read-only)." and provides a "Select repositories" button. Below that, it says "Selected 1 repository." and shows a list with "pradeepviswa/aws-devops-springbootapp" and a close button. Under "with these permissions:", there are two checked items: "Read access to actions, deployments, environments, issues, and metadata" and "Read and write access to administration, code, commit statuses, pull requests, and repository hooks". At the bottom, there are "Install & Authorize" and "Cancel" buttons. A note at the very bottom says "Next: you'll be redirected to https://redirect.codestar.aws/return".

13. Authorize



The screenshot shows a web browser window titled "Confirm access - Google Chrome". The address bar shows the URL "github.com/apps/aws-connector-for-github/installations". The page has a heading "Confirm access". Below it, it shows a user profile for "Signed in as @pradeepviswa". A message says "When your phone is ready, click the button below." with a green "Use GitHub Mobile" button. At the bottom, there is a section "Having problems?" with a link "Send a code via email".

14. Connect

The screenshot shows the AWS IAM console interface for creating a new connection to GitHub. The breadcrumb navigation is 'Developer Tools > Connections > Create connection'. The main heading is 'Connect to GitHub'. Below this is a 'GitHub connection settings' section with an 'Info' link. It contains a 'Connection name' field with the value 'AwsDevopsConnection'. Underneath is an 'App Installation - optional' section with a note: 'Install GitHub App to connect as a bot. Alternatively, leave it blank to connect as a GitHub user, which can be used in AWS CodeBuild projects.' There is a search field with the value '129386573'. At the bottom right is a large orange 'Connect' button.

15. Select newly created connection and save

The screenshot shows the 'Manage default source credential' page. The 'Source Provider' is set to 'GitHub'. Under 'Credential type', the 'GitHub App' option is selected, with a sub-note 'Connect project to GitHub using an AWS managed GitHub App'. Other options are 'Personal access token' and 'OAuth app'. The 'Connection' field shows a long ID: 'arn:aws:codeconnections:us-east-1:612990353408:connection/e7307754-cf2e-4be9-abda-60c0e02938e1'. Below this, the connection name 'AwsDevopsConnection' is listed. A 'Save' button is at the bottom right.

16. We get below message in "Create build project" screen and select repository

The screenshot shows the 'Create build project' screen. The 'Source' section is expanded, showing 'Source 1 - Primary'. The 'Source provider' is 'GitHub'. Under 'Credential', a green checkmark indicates 'Your account is successfully connected by using an AWS managed GitHub App. Manage account credentials.' There is an option to 'Use override credentials for this project only'. Under 'Repository', the 'Repository in my GitHub account' option is selected. The 'Repository' field contains the URL 'https://github.com/pradeepviswa/aws-devops-springbootapp'.

17. Let everything be default and scroll down to service role. Remember this role name as we will have to set permission

Service role

☒ New service role
Create a service role in your account

Role name

Type your service role name

18. Set Buildspec file

Buildspec

Build specifications

☐ Insert build commands
Store build commands as build project configuration

☒ Use a buildspec file
Store build commands in a YAML-formatted buildspec

Buildspec name - optional
By default, CodeBuild looks for a file named buildspec.yml in the source code root directory. If your buildspec file uses a different name or location, enter its path from the source root here (for example, buildspec.yml)

19. Click on **Create build project**

☐ S3 logs - optional
Checking this option will upload build output logs to S3.

[Cancel](#) [Create build project](#)

20. Project created

AwsDevopsProject

Configuration

Source provider	Primary repository	Artifacts upload location	Service role
GitHub	pradeepvishwa/aws-devops-springbootapp	-	arn:aws:iam::612990353408:role/service-role/AwsDevopsProject-service-role

Public builds
Disabled

[Build history](#) [Batch history](#) [Project details](#) [Build triggers](#) [Metrics](#) [Debug sessions](#)

[Actions](#) [Create trigger](#) [Edit](#) [Clone](#) [Debug build](#) [Start build with overrides](#) [Start build](#)

21. Go to iam -> roles -> select the role which created while create aws code build:.

click on: *AwsDevopsProject-service-role*

Identity and Access Management (IAM)

[Search IAM](#)

[Dashboard](#)

Access Management

[Roles](#)

Roles (1/7) [Info](#)

An IAM role is an identity you can create that has specific permissions with

[Search](#)

☒ **Role name**
[AwsDevopsProject-service-role](#)

☐ [AWSServiceRoleForCloudFormationStackSetsOrgMember](#)

22. Add Permission > attach policy

Permissions policies (1) [Info](#)

You can attach up to 10 managed policies.

Filter by Type: All types

☐	Policy name ↗	Type	Attached entities
☐	CodeBuildBasePolicy-AwsDevopsProject-service-role-us-east-1	Customer managed	1

[Add permissions](#) [Attach policies](#) [Create inline policy](#)

23. And below 3 access and click on “Add Permission”

- Administrator Access
- AmazonEC2ContainerRegistryFullAccess
- AmazonEC2ContainerRegistryPowerUser

Permissions policies (4) [Info](#)

You can attach up to 10 managed policies.

Filter by Type: All types

☐	Policy name ↗	Type	Attached entities
☐	AdministratorAccess	AWS managed - job function	3
☐	AmazonEC2ContainerRegistryFullAccess	AWS managed	1
☐	AmazonEC2ContainerRegistryPowerUser	AWS managed	1
☐	CodeBuildBasePolicy-AwsDevopsProject-service-role-us-east-1	Customer managed	1

24. Now go back to **Build Projects: “AwsDevopsProject”** and click on “Start Build” button

Success
Project details for build project AwsDevopsProject successfully updated.

AwsDevopsProject [Actions](#) [Create trigger](#) [Edit](#) [Clone](#) [Debug build](#) [Start build with overrides](#) [Start build](#)

Configuration

Source provider GitHub	Primary repository pradeepviwa/aws-devops-springbootapp	Artifacts upload location -	Service role arn:aws:iam::612990353408:role/service-role/AwsDevopsProject-service-role
Public builds Disabled			

25. Build Success

Build started
You have successfully started the following build: AwsDevopsProject:6ab1c93a-2570-4d9d-bba2-15c13026dee1

AwsDevopsProject:6ab1c93a-2570-4d9d-bba2-15c13026dee1 [Stop build](#) [Debug build](#) [Retry build](#)

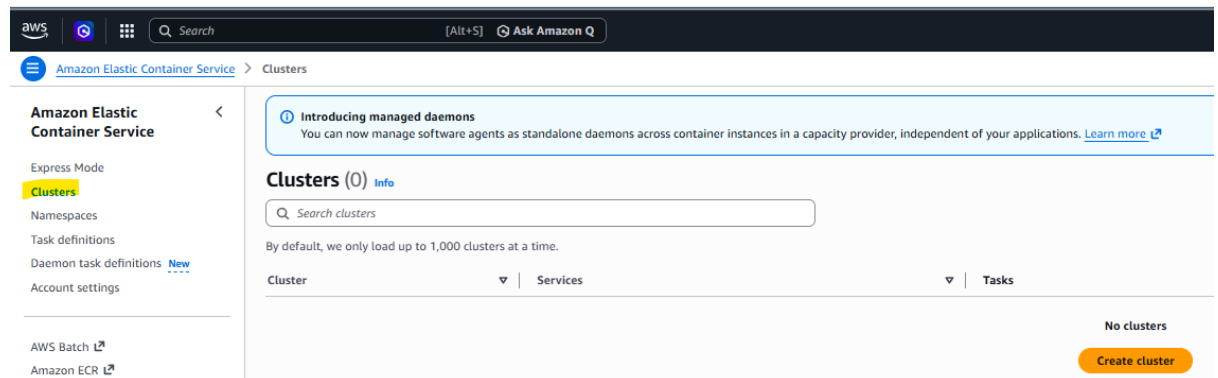
Build status

Status Succeeded	Initiator odi_user_2208052	Build ARN arn:aws:codebuild:us-east-1:612990353408:build/AwsDevopsProject:6ab1c93a-2570-4d9d-bba2-15c13026dee1	Resolved source version 320b5e853a043e05b47d063a2229a95511e9ca8c
Start time May 4, 2026 4:35 PM (UTC+5:30)	End time May 4, 2026 4:36 PM (UTC+5:30)	Build number 3	Reports AwsDevopsProject-JUNITXML-AutoDiscovered

[Build logs](#) [Phase details](#) [Reports](#) [Environment variables](#) [Build details](#) [Resource utilization](#)

26. Now we want to deploy the project, but we need to create ECS tasks and service.

Go to ecs -> create ecs



27. Cluster Name: AwsDevopsCluster-pv

Default namespace: AwsDevopsCluster-pv

Infrastructure: Fargate4 only

The screenshot shows the 'Create cluster' wizard in the AWS console. The title is 'Create cluster' with an 'Info' link. Below the title is a description: 'An Amazon ECS cluster groups together tasks, and services, and allows for shared cap'. The wizard is divided into sections. The first section is 'Cluster configuration'. It has a 'Cluster name' field with the value 'AwsDevopsCluster-pv'. Below this is a note: 'Cluster name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, hyphens (-), and u'. The next section is 'Service Connect defaults - optional'. It has a 'Default namespace' field with the value 'AwsDevopsCluster-pv'. Below this is a note: 'Select or type the namespace name to specify a group of services that make up your application'. The final section is 'Infrastructure - advanced' with an 'Info' link. It has a title 'Select a method of obtaining compute capacity' and a description: 'Your cluster is automatically configured for AWS Fargate (serverless), but you may choose to ac'. There are two radio button options: 'Fargate only' (which is selected) and 'EC2 instance profile'. The 'Fargate only' option has a description: 'Serverless - you don't think about creating or managing servers. Great for most common workloads.'

28. Cluster created

Clusters

Cluster AwsDevopsCluster-pv has been created successfully.

Introducing managed daemons
You can now manage software agents as standalone daemons across conta

Clusters (1) Info

Search clusters

By default, we only load up to 1,000 clusters at a time.

Cluster	Services
AwsDevopsCluster-pv	0

29. Click on Task definitions > create new task definition

Amazon Elastic Container Service

Express Mode
Clusters
Namespaces
Task definitions
Daemon task definitions
Account settings

Cluster AwsDevopsCluster-pv has been created successfully. View cluster

Task definitions (0) Info
Last updated: May 4, 2026, 16:48 (UTC+5:30) | Display | Create new revision | Create new task definition

Filter task definitions | Filter status: Active

Task definition | Status of last revision

No task definitions
No task definitions to display.
Create new task definition

30. Task definition name: AwsDevopsTaskDefinition Infrastructure: Fargate

Cluster AwsDevopsCluster-pv has been created successfully.

Create new task definition Info

Task definition configuration

Task definition family Info
Specify a unique task definition family name.

AwsDevopsTaskDefinition

Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.

▼ **Infrastructure requirements** Info
Specify the infrastructure requirements for the task definition.

Launch type Info
Selection of the launch type will change task definition parameters.

☒ **AWS Fargate**
Serverless compute for containers.

- ▼ Container - 1 [Info](#)

Container details

Specify a name, container image, and whether the container should be marked as essential. Each task definition must have at least one essential container.

Name

Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.

Essential container

Yes

▼

Image URI

Up to 255 letters (uppercase and lowercase), numbers, hyphens, underscores, colons, periods, forward slashes, and number signs are allowed.

- Port mappings

Info

Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.

Container port	Protocol	Port name	App protocol	
8080	TCP	container-port-protocol	HTTP	Remove

Add port mapping

- Task definition successfully created

AwsDevOpsTaskDefinition:1 has been successfully created. You can use this task definition to deploy a service or run a task.

View task definition

Notifications

AwsDevOpsTaskDefinition:1

Last updated

May 4, 2026, 17:54 (UTC+5:30)

Deploy

Actions

Create new revision

Overview

ARN

aws:ecs:us-east-1:612990353408:task-definitions/AwsDevOpsTaskDefinition:1

Task role

-

Fault injection

Turned off

Status

ACTIVE

Task execution role

[ecsTaskExecutionRole](#)

Time created

May 4, 2026, 17:54 (UTC+5:30)

Operating system/Architecture

Linux/X86_64

App environment

Fargate

Network mode

awsvc

Containers

JSON

Task placement

Volumes (0)

Requires attributes

Tags

- Amazon Elastic Container Service

Clusters

Express Mode

Clusters

Namespaces

Task definitions

Daemon task definitions New

Account settings

AWS Batch

Amazon ECR

Repositories

Task definition successfully created

AwsDevopsTaskDefinition:1 has been successfully created. You can now create a cluster with this task definition.

Introducing managed daemons

You can now manage software agents as standalone daemons.

Clusters (1) Info

Q Search clusters

By default, we only load up to 1,000 clusters at a time.

Cluster	Services
AwsDevopsCluster-pv	0

35. GO TO ECS -> CLUSTER -> CLICK ON THE CLUSTER WE CREATD
will create a service. Click on **create**

AwsDevopsCluster-pv

Cluster overview

ARN  arn:aws:ecs:us-east-1:612990353408:cluster/AwsDevopsCluster-pv	Status  Active	CloudWatch logs  Def
Services Draining -	Active -	Tasks Pending -

Services

Daemons New

Tasks

Infrastructure

Metrics

Scheduled tasks

Configuration

Event history

Tags

Services (0) [Info](#)

Filter launch type
Any launch type

Filter scheduling strategy
Any scheduling strategy

☐	Service name	▲	ARN	Status	▼	Scheduling strategy	▼	Launch type
No services								
No services to display.								
Create								

36. Select **Task definition** and **Compute configuration** as below
keep any service name

> Create service

Task definition family
Select an existing task definition family. To create a new task definition, go to [Task definitions](#).

AwsDevopsTaskDefinition

Task definition revision Latest
Select the task definition revision from the 100 most recent entries, or enter a revision. Leave the field blank to use the latest revision.

1

Service name
Assign a service name that is unique for this cluster.

AwsDevopsTaskDefinition-service-m0s87ej9

Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed. Service names must be unique within a cluster.

Environment

Existing cluster

AwsDevopsCluster-pv

▼ **Compute configuration - advanced**

Compute options | [Info](#)
To ensure task distribution across your compute types, use appropriate compute options.

☐ **Capacity provider strategy**
Specify a launch strategy to distribute your tasks across one or more capacity providers.

☒ **Launch type**
Launch tasks directly without the use of a capacity provider strategy.

Launch type | [Info](#)
Select either managed capacity (Fargate), or custom capacity (EC2 or user-managed, External instances). External instances are registered to your cluster using the ECS Anywhere capability.

Fargate

Platform version | [Info](#)
Specify the platform version on which to run your service.

LATEST

► **Troubleshooting configuration - recommended**

Deployment configuration

Scheduling strategy | [Info](#)

☒ **Replica**
Place and maintain a desired number of tasks across your cluster.

☐ **Daemon**
Place and maintain one copy of your task on each container instance.

37. Service created

AwsDevopsCluster-pv

Last updated May 4, 2026, 18:03 (UTC+5:30) [Actions](#) [Create with Express Mode](#)

Cluster overview

ARN arn:aws:ecs:us-east-1:612990553408:cluster/AwsDevopsCluster-pv	Status Active	CloudWatch monitoring Default	Registered container instances -
Services Draining -	Active 1	Tasks Pending -	Running 1

[Services](#) [Daemons](#) [Tasks](#) [Infrastructure](#) [Metrics](#) [Scheduled tasks](#) [Configuration](#) [Event history](#) [Tags](#)

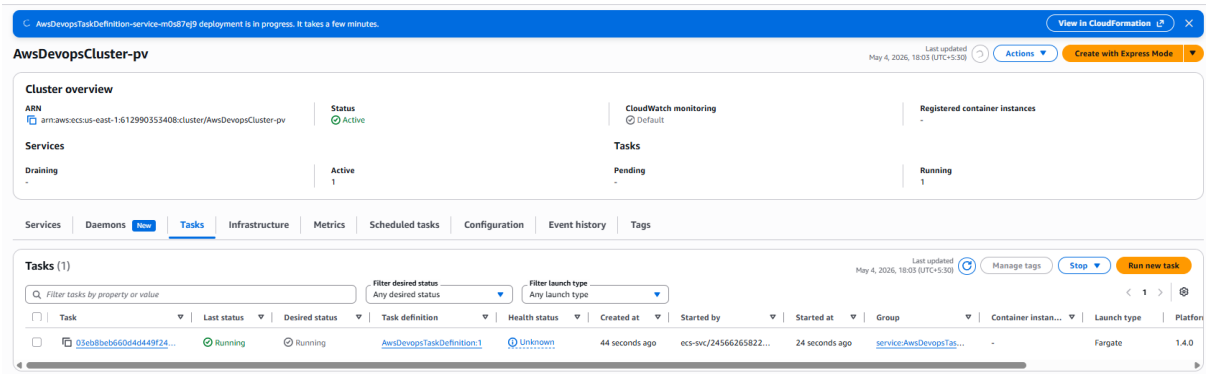
Services (1) [Info](#)

Last updated May 4, 2026, 18:03 (UTC+5:30) [Manage tags](#) [Update](#) [Delete service](#) [Create](#)

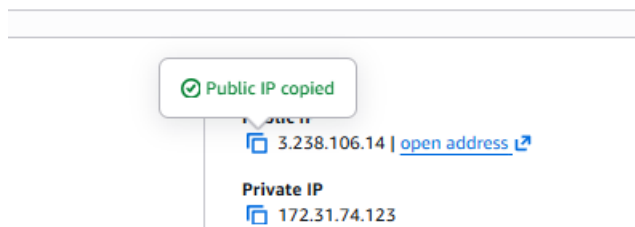
Filter services by value

Service name	ARN	Status	Scheduling strategy	Launch type	Task definition	Deployments and tasks	Last deployment	Created at
AwsDevopsTaskDefinition-service-m0s...	arn:aws:ecs:us-east-1:612990553408:task-definition/AwsDevopsTaskDefinition:1	Active	Replica	Fargate	AwsDevopsTaskDefinition:1	0/1 Tasks running	In progress	25 seconds ago

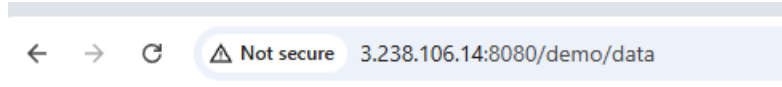
38. Task created, click on task



39. Copy public ip

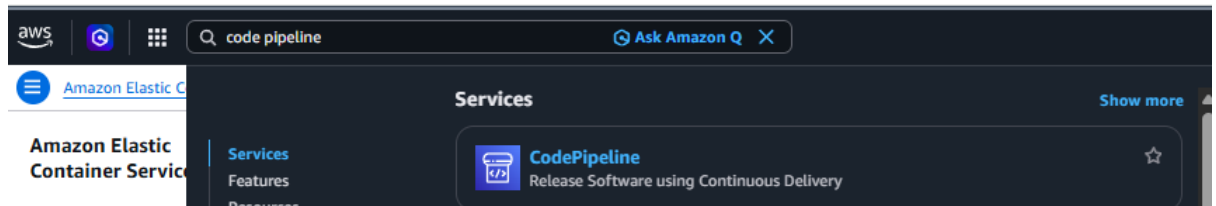


40. Open browser and open path like this: http://3.238.106.14:8080/demo/data

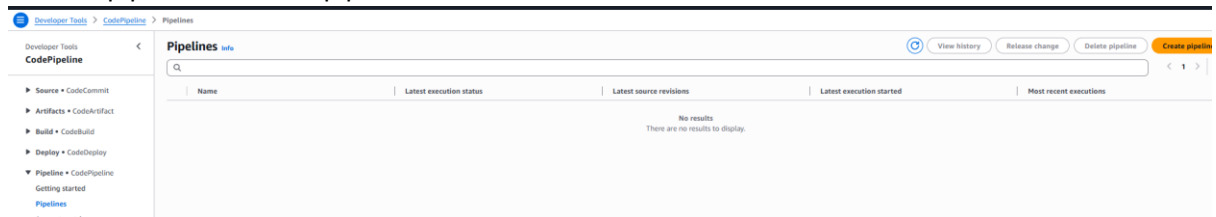


First message from AWS Ecs

41. Now we will build the code pipeline and automate everything in code pipeline



42. Click on pipelines > create pipeline



43. Build custom pipeline > next

Step 1

Choose creation option

Step 2

Choose pipeline settings

Step 3

Add source stage

Step 4 - optional

Add build stage

Step 5 - optional

Add test stage

Step 6

Add deploy stage

Step 7

Review

Choose creation option [Info](#)

Step 1 of 7

Category

☐ Deployment

☐ Continuous Integration

☐ Automation

☒ Build custom pipeline

[Cancel](#)[Next](#)

44. Pipeline name: AwsDevopsPipeline

Step 1

[Choose creation option](#)

Step 2

Choose pipeline settings

Step 3

Add source stage

Step 4 - optional

Add build stage

Step 5 - optional

Add test stage

Step 6

Add deploy stage

Step 7

Review

Choose pipeline settings [Info](#)

Step 2 of 7

Pipeline settings

Pipeline name
Enter the pipeline name. You cannot edit the pipeline name after it is created.

AwsDevopsPipeline

No more than 100 characters

Execution mode [Info](#)
Choose the execution mode for your pipeline. This determines how the pipeline is run.

☐ Superseded

☒ Queued

☐ Parallel

Service role

☒ New service role
Create a service role in your account

☐ Existing service role
Choose an existing service role from your account

Role name

AWSCodePipelineServiceRole-us-east-1-AwsDevopsPipeline

Type your service role name

☒ Allow AWS CodePipeline to create a service role so it can be used with this new pipeline

► **Advanced settings**
Configure artifact store location, encryption settings, and pipeline variables for your pipeline.

[Cancel](#)[Previous](#)[Next](#)

45. Source provider: GitHub (via OAuth app)

Connect to github > A popup will appear > Select authorise code suite > Connect

Step 1

Choose creation option

Step 2

Choose pipeline settings

Step 3

Add source stage

Step 4 - optional

Add build stage

Step 5 - optional

Add test stage

Step 6

Add deploy stage

Step 7

Review

Add source stage Info

Step 3 of 7

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (via OAuth app) ▼

Grant AWS CodePipeline access to your GitHub repository. This allows AWS CodePipeline to upload commits from GitHub to your pipeline.

Connected

✔ You have successfully authenticated your account. ✕

ⓘ The GitHub (via OAuth app) action is not recommended

The selected action uses OAuth apps to access your GitHub repository. This is no longer the recommended method. Instead, choose the GitHub (via GitHub App) action to access your repository by creating a connection. Connections use GitHub Apps to manage authentication and can be shared with other resources. [Learn more](#) ⓘ

Repository

Q pradeepviswa/aws-devops-springbootapp ✕

Branch

Q master ✕

☒ Enable automatic retry on stage failure

Cancel

Previous

Next

46. Click on **other build providers**

Step 1

Choose creation option

Step 2

Choose pipeline settings

Step 3

Add source stage

Step 4 - optional

Add build stage

Step 5 - optional

Add test stage

Step 6

Add deploy stage

Step 7

Review

Add build stage - optional

Step 4 of 7

Build - optional

Build provider

Choose the tool you want to use to run build commands and specify artifacts for your build action.

Commands

Other build providers

AWS CodeBuild

Project name

Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

Q AwsDevopsProject X

or

Create project ↗

☐

Define buildspec override - optional

Buildspec file or definition that overrides the latest one defined in the build project, for this build only.

Environment variables - optional

Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. [Learn more ↗](#)

Add environment variable

Build type

Single build

Triggers a single build.

Batch build

Triggers multiple builds as a single execution.

Region

United States (N. Virginia)

Input artifacts

Choose an input artifact for this action. [Learn more ↗](#)

SourceArtifact X

Defined by: Source

☒

Enable automatic retry on stage failure

Cancel

Previous

Skip build stage

Next

47. Step 5

Step 1

Choose creation option

Step 2

Choose pipeline settings

Step 3

Add source stage

Step 4 - optional

Add build stage

Step 5 - optional

Add test stage

Step 6

Add deploy stage

Step 7

Review

Add test stage - optional [Info](#)

Step 5 of 7

Test - optional

Test provider
Choose how you want to test your application or content. Choose the provider, and then provide the configuration details for that provider.

AWS CodeBuild

Region
United States (N. Virginia)

Input artifacts
Choose an input artifact for this action. [Learn more](#)

BuildArtifact
Defined by: Build

No more than 100 characters

Project name
Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

AwsDevopsProject or Create project

Use: "AwsDevopsProject"

AwsDevopsProject

one defined in the build project, for this build only.

Environment variables - optional
Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. [Learn more](#)

Add environment variable

Build type

☒ Single build
Triggers a single build.

☐ Batch build
Triggers multiple builds as a single execution.

☒ Enable automatic retry on stage failure

[Cancel](#) [Previous](#) [Skip test stage](#) [Next](#)

48. Step 7

deploy provider: Amazon ECS

Step 1

Choose creation option

Step 2

Choose pipeline settings

Step 3

Add source stage

Step 4 - optional

Add build stage

Step 5 - optional

Add test stage

Step 6

Add deploy stage

Step 7

Review

Add deploy stage

Info

Step 6 of 7

Deploy - optional

Deploy provider

Choose how you want to deploy your application or content. Choose the provider, and then provide the configuration details for that provider.

Amazon ECS

Region

United States (N. Virginia)

Input artifacts

Choose an input artifact for this action. [Learn more](#)

BuildArtifact

×

Defined by: Build

No more than 100 characters

Cluster name

Choose a cluster that you have already created in the Amazon ECS console. Or create a cluster in the Amazon ECS console and then return to this task.

AwsDevopsCluster-pv

×

Service name

Choose a service that you have already created in the Amazon ECS console for your cluster. Or create a new service in the Amazon ECS console and then return to this task.

AwsDevopsTaskDefinition-service-m0s87ej9

×

Image definitions file - optional

Enter the JSON file that describes your service's container name and the image and tag.

MyFilename.json

Deployment timeout - optional

Enter the timeout in minutes for the deployment action.

☒ Configure automatic rollback on stage failure

☐ Enable automatic retry on stage failure

Cancel

Previous

Skip deploy stage

Next

49. Create pipeline

Step 6: Add deploy stage

Deploy action provider

Deploy action provider

Amazon ECS

ClusterName

AwsDevopsCluster-pv

ServiceName

AwsDevopsTaskDefinition-service-m0s87ej9

Configure automatic rollback on stage failure

Enabled

Enable automatic retry on stage failure

Disabled

Cancel

Previous

Create pipeline

50. Pipeline Created, just remove test stage and run the pipeline

Developer Tools > CodePipeline > Pipelines > AwsDevopsPipeline

AwsDevopsPipeline

Pipeline

Executions

Triggers

Settings

Tags

Stage

Source

a7d784e4-7ec2-4ea4-858d-676cde749bc3

All actions succeeded.

Source

GitHub (via OAuth app)

May 4, 2026 6:50 PM (UTC+5:30)

320b5e85 Source: Update Dockerfile

Build

a7d784e4-7ec2-4ea4-858d-676cde749bc3

All actions succeeded.

Build

AWS CodeBuild

May 4, 2026 6:51 PM (UTC+5:30)

320b5e85 Source: Update Dockerfile

Deploy

a7d784e4-7ec2-4ea4-858d-676cde749bc3

All actions succeeded.

Deploy

Amazon ECS

May 4, 2026 6:55 PM (UTC+5:30)

320b5e85 Source: Update Dockerfile

✓

✓

✓

ⓘ